Task1:
In this task, I tested various sizes of chunks and settings of overlap to determine the effect of the retrieval performance. I tested (256, 0), (128, 0), (256, 50), (256, 100), (512, 0), (512, 50), and (512, 100). I also compared sentence based splitting and paragraph based splitting. After multiple run and evaluating the retrieval metrics I found that using paragraph-based splitting with a chunk size of (512, 100) consistently produced better results than the other configurations, since the medical data in the WebMD document was never contained in single sentences but in meaningful and structured paragraphs.

Out of all the configurations, (512, 100) was always reporting higher retrieval results. Smaller chunk size at times fragmented key medical explanations into partial ones, hence minimized relevancy. At overlap 0, there was loss of some key information at boundaries. A further increase in overlap to 100 assisted in maintaining context between chunks. Since (512, 100) yielded higher values of MRR and hit rate than all other combinations.

I picked it as my final ingestion setting and used as the point of reference of the rest of the experiments. Before even changing any retrieval algorithms it has showed a difference in the output along with the pdf cleaning techniques.

Task2:
In task 2, I developed a varied list of questions to assess the advantages and disadvantages of various retrieval strategies. Alternatively to simple factual questions, I had cross-sectional ones necessitating the retrieval of the information across various sections, key-word-intensive questions necessitating the use of exact medical terminology, semantic ones that involved knowledge of the context, and structured-data questions, which involved the retrieval of information in lists or formatted sections. This assisted me in seeing how the performance of the three types of queries behaved with the use of the vector-based retrieval, BM25, and auto-merging.

In my experimentation, I observed obvious trade-offs. BM25 was more effective with keyword intensive queries in which terms were crucial,
and the semantic or contextual questions were better handled by the use of vectors. Auto-Merging Retriever had a tendency to work better
on cross-sectional questions since the relevant chunks were clustered together and the context was maintained. The creation of this test
set allowed me to realize more clearly that the performance of the test depends on the type of questions significantly, and there is no
universally good retrieval strategy.

Task3:
In this task, I noted that not everything that is important in the WebMD PDF was being reflected during the normal text extraction. Certain information was included in images and formatted visuals and this was totally disregarded by the original ingestion pipeline. To correct this, I have used an OCR tool to get text out of those images. I put everything that was extracted in one document and added it to the existing corpus then ran the ingestion pipeline once more.

Once the OCR was integrated, I reran the retrieval queries and then compared the results with the old version. I also observed that when it came to structured or visually presented information, especially on questions that were the most structured, the answer provided became more detailed. Prior to the OCR integration, the system occasionally used to produce fragmented responses, due to the impossibility of detecting the content within the images. That text inclusion improved the coverage of the system over the document and allowed the system

to fetch a more relevant context.Images can conceal important information and when OCR is not applied, the RAG system will be deprived of such information. Introduction of OCR did not radically alter all the metrics, although it increased overall completeness and facilitated the retrieval system.

Task4:
In this task, I used hybrid retrieval system where I used a weighted fusion of vector-based retrieval system and BM25. I used different retrievers in each of the methods with a greater top-k value and then fused them together using score-based fusion (biasing more heavily towards the vector retriever but still using BM25 signals). The idea was to take advantage of each of the two methods- semantic understanding as provided by the embeddings and the accuracy of the keyword matching as provided by BM25.

Having reviewed the hybrid retriever, I noted that it usually worked more reliably across question type than the use of either strategy separately. BM25 is useful in terms of queries that are heavy with keywords, whereas semantic and contextual queries worked better with the use of the vector search. The hybrid strategy minimized the weaknesses observed in the individual retrievers; and enhanced the balance in retrieval. In this experiment, it was found that hybrid retrieval is stronger, particularly in the process of answering various question types.

Task5:
In this task, I focused to retrieval precision and quality ranking focusing on refining the retrieval phase. During the previous experiments I noticed that, though in many instances they were relevant chunks they still appeared not to be ranked at the top positions, thus influencing completeness of answers. My improvement led to quantifiable improvement in ranking-based metrics. In particular,MRR went up to approximately 0.13 to about 0.18, and hit rate went up to 0.23 to approximately 0.29. There was also a slight improvement in precision which means that more relevant chunks were being shown earlier in the rankings.

The responses were more focused and did not need as much irrelevant context qualitatively. Cross-sectional and semantic queries were more consistent across the various types of questions in the system. Although the performance gains were not dramatic, but rather linear, it shows that when refinements are made to the retrieval pipeline to focus the overall system performance, they can contribute significantly to the performance of the system.

This assignment confirmed that streamlining a RAG system is not a big change but the critical identification of bottlenecks and their enhancement step-by-step with the help of both quantitative and qualitative assessment.